Literature Survey and Algorithmic Analysis for Real-Time EOG-Based Control Systems

1 Introduction

Electrooculography (EOG) is a non-invasive technique that captures the corneoretinal standing potential existing between the front (cornea) and back (retina) of the human eye. As the eye rotates, this dipole shifts, generating distinct electrical signals that can be acquired using surface electrodes. Our proposed project implements a comprehensive real-time hardware and software pipeline for EOG-based control. The system utilizes BioAmp electrodes interfaced with an ADS1115 analog-to-digital converter (ADC), which transmits digitized signals via an Arduino over a USB-serial connection.

To process the continuous data stream, our Python-based pipeline applies causal digital filtering, sliding-window segmentation, and normalization. For decision-making, the system decouples blink events from directional eye movements. It employs a dedicated heuristic thresholding detector equipped with debounce and refractory logic for blink and triple-blink identification. Concurrently, directional gazes (Left, Right, Up, Down) are classified using a lightweight machine learning classifier operating on time-domain features. Finally, a custom temporal state machine stabilizes the noisy predictions to drive a real-time graphical user interface (GUI). While the software architecture has been thoroughly validated on a custom synthetic data simulator, final validation on real EOG hardware signals remains a documented future step.

2 Literature Survey

A review of recent literature reveals diverse algorithmic approaches for processing EOG signals, ranging from simple thresholding to complex deep learning architectures.

Hardware-Level Thresholding for Drowsiness Detection

Jambhekar and Ram (2017) proposed an EOG acquisition system utilizing a PIC Microcontroller to detect driver drowsiness [1]. The system extracts the amplitude and frequency of the EOG signal directly. By establishing predefined amplitude thresholds, the algorithm detects eye blinks. A drowsy state is triggered if the system counts five consecutive prolonged blinks. This study demonstrates that simple heuristic thresholding is highly effective for event detection on microcontrollers with extremely low computational overhead.

Time-Frequency Representation and CNN Classification

Zyout et al. (2023) introduced a hybrid deep learning approach to classify five EOG movements (Blink, Horizontal Left, Horizontal Right, Vertical Down, Vertical Up) [2].

The authors transformed the 1D EOG time-series data into 2D circular spectrograms known as "Irisgrams". A pre-trained Convolutional Neural Network (ResNet101) was employed to automatically extract deep features from these images, which were then passed to a Support Vector Machine (SVM). This computationally intensive approach achieved a high test accuracy of 96.2%.

Machine Learning and Deep Learning for Robotic Arm Control

Alshehhi et al. (2025) evaluated multiple learning architectures to translate EOG signals into control commands for a robotic arm [3]. Utilizing the BNCI Horizon 2020 dataset, they tested SVM, CNN, and Long Short-Term Memory (LSTM) networks. The SVM model, trained on time-domain features (mean, standard deviation, energy, zero-crossing rate, skewness, and kurtosis), achieved an accuracy of 82.0%. A 1D-CNN operating on raw segments achieved 90.3%, while the LSTM architecture, which inherently captures temporal dependencies in sequential data, achieved the highest accuracy at 93.2%.

3 Algorithmic Comparison

A. Threshold / Heuristic Methods

- **Mechanism:** Analyzes the raw signal's amplitude and frequency against predefined limits to detect specific physiological events.
- **Advantages & Limitations:** This method boasts near-zero latency and minimal computational cost, making it ideal for embedded microcontrollers. However, it is fundamentally limited to detecting distinct spikes (like blinks) and cannot reliably classify complex, multi-directional continuous gaze vectors.
- **Real-Time Suitability:** Highly suitable for instantaneous event triggering.

B. Traditional Machine Learning (LDA & SVM)

- **Mechanism:** These algorithms rely on manual feature extraction. Time-domain statistics (such as mean, variance, and signal energy) are extracted from segmented windows. SVM separates classes by mapping features into higher-dimensional spaces using kernel functions (e.g., RBF), while algorithms like Linear Discriminant Analysis (LDA) project data to maximize class separability.
- **Advantages & Limitations:** They require significantly less computational power than neural networks and allow for rapid, person-specific calibration. While accuracy (e.g., 82.0% for SVM) may trail behind deep learning models, their inference speed is optimal for live streaming.
- **Real-Time Suitability:** Highly suitable for processing live data streams on standard computing hardware.

C. Long Short-Term Memory (LSTM)

- **Mechanism:** LSTMs are specialized recurrent neural networks designed to learn long-term temporal dependencies directly from raw sequential data.
- **Advantages & Limitations:** LSTM yields high classification accuracy (93.2%) for continuous EOG sequences. However, it requires large datasets for training and entails heavy matrix multiplications during inference, potentially introducing latency in resource-constrained environments.

D. Convolutional Neural Networks (CNN)

- **Mechanism:** CNNs automatically learn hierarchical spatial features. In EOG applications, raw signals are often transformed into 2D time-frequency representations (like spectrograms or Irisgrams) before classification.
- **Advantages & Limitations:** CNNs achieve state-of-the-art accuracy (up to 96.2%). However, the preprocessing overhead of converting real-time 1D streams into 2D images, coupled with deep feature extraction, makes inference computationally expensive and difficult to execute in a low-latency control loop.

4 Existing Approaches vs. Proposed System

Table 1: Algorithm Comparison across Literature and Proposed System

Research Work / Algorithm	Input Data	Feature Extraction	Classifier	Accuracy / Result	Computational Load	Real-Time Suitability	Relation to Our Project
Jambhekar et al. (2017)	Raw EOG	None (Amplitude)	Thresholding	Not reported (alarm based)	Very Low	High	Project employs similar low-latency heuristic logic for blink detection.
Alshehhi et al. (2025)	Normalized EOG	Time-domain	SVM	82.0%	Low	High	Validates the use of time-domain features for real-time directional classification.
Alshehhi et al. (2025)	Raw EOG	Automatic (Sequential)	LSTM	93.2%	High	Low (on limited hardware)	Highlights that sequential deep models carry heavy computational requirements.
Zyout et al. (2023)	Irisgram (2D)	Pre-trained ResNet101	CNN + SVM	96.2%	Very High	Very Low	Excluded from our project due to the heavy image transformation overhead.
Our Proposed System (*eog_ball_control*)	Streaming EOG	Sliding Window Time-Domain	Hybrid (Heuristic + ML)	Validated on Synthetic Simulator	Low	Very High	Demonstrates an end-to-end synchronous pipeline with temporal stabilization.

Table 2: Design Parameter Comparison

Parameter	**Existing EOG Approaches**	**Our Proposed System (*eog_ball_control*)**
Signal Acquisition	Mostly offline pre-recorded datasets	Live hardware (ADS1115 via Arduino USB-serial) & Simulator
Feature Extraction	Deep learned features or Irisgrams	Computational-light Time-Domain statistics
Direction Classification	Heavy deep learning architectures	Lightweight ML classification
Blink Detection	Often merged into main ML model	Independent heuristic detector with debounce logic
Temporal Stabilization	Frequently omitted	Custom Temporal State Machine (filters noisy/low-confidence outputs)
Computational Requirement	High (Deep learning inference)	Low (Bounded queues, threaded pipeline)
Main Limitation	Lacks live hardware-to-software integration	Awaiting full validation on real EOG hardware signals

5 Algorithm Selection Rationale for the Proposed System

The primary objective of the proposed system is to maintain a continuous, low-latency control loop that translates hardware EOG signals into real-time GUI actions. Consequently, a hybrid algorithmic approach was selected.

Why Thresholding is used for Blink Detection

A blink is a distinct, short-duration physiological event rather than a continuous directional gaze. Attempting to force blink detection into a continuous gaze classifier often increases false positives. Therefore, our system implements a dedicated heuristic blink detector. By applying specific thresholds alongside debounce and refractory windows, the logic accurately isolates single and triple blinks (used as "stop" commands) with zero processing latency.

Why Lightweight Machine Learning is used for Directional Gaze

For classifying continuous ocular movements (Left, Right, Up, Down), the project extracts low-dimensional time-domain features over sliding windows. Lightweight machine learning classifiers (such as LDA or SVM) are inherently suited for this task. They require significantly smaller datasets for subject-specific calibration and execute inference in sub-milliseconds. This ensures that the serial data stream from the Arduino is not bottlenecked and the Tkinter GUI remains responsive.

Why CNN and LSTM are currently excluded

While deep learning models yield superior accuracy, their architectural demands conflict with the project's real-time hardware constraints. Generating 2D Irisgrams for a CNN introduces severe preprocessing delays. Similarly, an LSTM requires recurrent matrix computations over sequential buffers. Processing these heavy models in real-time alongside causal filters, continuous serial reads, and GUI rendering would disrupt the synchronous pipeline, leading to dropped frames and queue overflows.

6 Research Gap

Existing research predominantly focuses on maximizing classification accuracy using offline benchmark datasets and computationally heavy deep learning models (CNNs, LSTMs). These studies often assume ideal signal conditions and lack an integrated, real-world deployment pipeline.

The significant research gap lies in the absence of complete, end-to-end hardware-to-software integration intended for live control. Our project addresses this gap by developing a synchronized pipeline that combines serial acquisition, digital filtering, independent blink heuristics, real-time ML inference, and crucially, a temporal state machine. This temporal stabilization is vital for filtering out transient noise and suppressing gaze predictions during blinks, providing a robust interface for actual human-computer interaction.

7 Conclusion

Developing a real-time EOG-based control system requires a careful balance between classification accuracy and computational latency. While deep neural networks (CNN, LSTM) offer high theoretical accuracy on offline datasets, their processing overhead is prohibitive for low-latency live streaming. The proposed system effectively bridges the gap between hardware acquisition and software execution by employing a hybrid architecture: applying rapid heuristic logic for blink events and lightweight time-domain machine learning for continuous directional classification. This structural choice ensures a stable, responsive, and robust control loop suitable for assistive interactive technologies.

References

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[3] M. Alshehhi, A. Singh, and J. Mounsef, "Eye Gesture-Based Robotic Arm Control Using EOG and Machine Learning," in *2025 2nd International Conference on Artificial Intelligence, Metaverse, and Cybersecurity (ICAMAC)*, 2025, pp. 1-6.